MODULE – 1

INTRODUCTION

Introduction:

Research comprises defining and redefining problems, formulating hypothesis or suggested solution; collecting, organizing and evaluating data, making deductions and reaching conclusions and carefully testing the conclusions to determine whether they fit the formulating hypothesis. The manipulation of things, concepts or symbols for the purpose of generalizing to extend, correct or verify knowledge, whether that knowledge aids in construction of theory or in the practice of an art.

Research in simple terms refers to search for knowledge. It is a scientific and systematic search for information on a particular topic or issue. It is also known as the art of scientific investigation. Several social scientists have defined research in different ways.

In the *Encyclopedia of Social Sciences*, D. Slesinger and M. Stephension (1930) defined research as "the manipulation of things, concepts or symbols for the purpose of generalizing to extend, correct or verify knowledge, whether that knowledge aids in the construction of theory or in the practice of an art".

What is the definition of research methodology?

The process used to collect information and data for the purpose of making business decisions. The methodology may include publication research, interviews, surveys and other research techniques, and could include both present and historical information. Research methodology is a term that basically means the science of how research is done scientifically. It is a way to systematically and logically solve a problem, help us understand the process not just the product of research, and analyzes methods in addition to the information obtained by them.

What are the types of research methodology?

- Basic research
- Applied Research
- Problem oriented research
- Problem solving
- Quantitative Research
- Qualitative Research

Research Methods Vs Methodology:

Research methods include all those techniques/methods that are adopted for conducting research. Thus, research techniques or methods are the methods that the researchers adopt for conducting the research studies. On the other hand, research methodology is the way in which research problems are solved systematically. It is a science of studying how research is conducted scientifically. Under it, the researcher acquaints himself/herself with the various steps generally adopted to study a research problem, along with the underlying logic behind them. Hence, itis not only important for the researcher to know the research techniques/methods, but also the scientific approach called methodology.

Table: Methods vs Techniques

Type	Methods	Techniques
Library Research	(i) Analysis of historical	Recording of notes, Content analysis, Tape and Film listening and Research records analysis.
	(ii) Analysis of documents	Statistical compilations and manipulations, reference and abstract guides, contents analysis.
Field Research	(i) Non-participant direct observation	Observational behavioural scales, use of score cards, etc.
	(ii) Participant observation	Interactional recording, possible use of tape recorders, photo graphic techniques.
	(iii) Mass observation	Recording mass behaviour, interview using independent observers in public places.
	(iv) Mail questionnaire	Identification of social and economic background of respondents.
	(v) Opinionnaire	Use of attitude scales, projective techniques, use of sociometric scales.
	(vi) Personal interview	Interviewer uses a detailed schedule with open and closed questions.
	(vii) Focused interview	Interviewer focuses attention upon a given experience and its effects.
	(viii) Group interview	Small groups of respondents are interviewed simultaneously.
	(ix) Telephone survey	Used as a survey technique for information and for discerning opinion; may also be used as a follow up of questionnaire.
	(x) Case study and life history	Cross sectional collection of data for intensive analysis, longitudinal collection of data of intensive character.
Laboratory Research	Small group study of random behaviour, play and role analysis	Use of audio-visual recording devices, use of observers, etc.

Meaning of Research:

Research in common parlance refers to a search for knowledge. Once can also define research as a scientific and systematic search for pertinent information on a specific topic. In fact, research is an art of scientific investigation. The Advanced Learner's Dictionary of Current English lays down the meaning of research as "a careful investigation or inquiry specially through search for new facts in any branch of knowledge."1 Redman and Mory define research as a "systematized effort to gain new knowledge."2 Some people consider research as a movement, a movement from the known to the unknown. It is actually a voyage of discovery. We all possess the vital instinct of inquisitiveness for, when the unknown confronts us,we wonder and our inquisitiveness makes us probe and attain full and fuller understanding of the unknown. This inquisitiveness is the mother of all knowledge and the method, which man employs for obtaining the knowledge of whatever the unknown, can be termed as research.

Research is an academic activity and as such the term should be used in a technical sense. According to Clifford Woody research comprises defining and redefining problems, formulating hypothesis or suggested solutions; collecting, organising and evaluating data; making deductions and reachingconclusions; and at last carefully testing the conclusions to determine whether they fit the formulating hypothesis. D. Slesinger and M. Stephenson in the Encyclopaedia of Social Sciences define research as "the manipulation of things, concepts or symbols for the purpose of generalising to extend, correct or verify knowledge, whether that knowledge aids in construction of theory or in the practice of an art."3 Research is, thus, an original contribution to the existing stock of knowledge making for its advancement. It is the persuitof truth with the help of study, observation, comparison and experiment. In short, the search for knowledge through objective and systematic method of finding solution to a problem is research. The systematic approach concerning generalisation and the formulation of a theory is also research. As such the term 'research' refers to the systematic method consisting of enunciating the problem, formulating a hypothesis, collecting the facts or data, analysing the facts and reaching certain conclusions either in the form of solutions(s) towards the concerned problem or in certain generalisations for some theoretical formulation.

Objectives of Engineering Research:

The purpose of research is to discover answers to questions through the application of scientific procedures. The main aim of research is to find out the truth which is hidden and which has not been discovered as yet. Though each research study has its own specific purpose, we may think of research objectives as falling into a number of following broad groupings:

1. To gain familiarity with a phenomenon or to achieve new insights into it (studies with this object in view are termed as *exploratory* or *formulative* research studies);

2. To portray accurately the characteristics of a particular individual, situation or a group (studies with this object in view are known as *descriptive* research studies);

3. To determine the frequency with which something occurs or with which it is associated with something else (studies with this object in view are known as *diagnostic* research studies);

4. To test a hypothesis of a causal relationship between variables (such studies are known as *hypothesis-testing* research studies).

Motivation in Engineering Research:

What makes people to undertake research? This is a question of fundamental importance. The possible motives for doing research may be either one or more of the following:

1. Desire to get a research degree along with its consequential benefits;

2. Desire to face the challenge in solving the unsolved problems, i.e., concern over practical problems initiates research;

3. Desire to get intellectual joy of doing some creative work;

4. Desire to be of service to society;

5. Desire to get respectability.

However, this is not an exhaustive list of factors motivating people to undertake research studies. Many more factors such as directives of government, employment conditions, curiosity about new things, desire to understand causal relationships, social thinking and awakening, and the like may as well motivate (or at times compel) people to perform research operations.

Types of Engineering Research:

The basic types of research are as follows:

(i) ***Descriptive vs. Analytical:*** *Descriptive research* includes surveys and fact-finding enquiries of different kinds. The major purpose of descriptive research is description of the state of affairs as it exists at present. In social science and business research we quite often use the term *Ex post facto research* for descriptive research studies. The main characteristic of this method is that the researcher has no control over the variables; he can only report what has happened or what is happening. Most *ex post facto research* projects are used for descriptive studies in which the researcher seeks to measure such items as, for example, frequency of shopping, preferences of people, or similar data. *Ex post facto studies* also include attempts by researchers to discover causes even when they cannot control the variables. The methods of research utilized in descriptive research are survey methods of all kinds, including comparative and correlational methods. In *analytical research*, on the other hand, the researcher has to use facts or information already available, and analyze these to make a critical evaluation of the material.

(ii) ***Applied vs. Fundamental:*** Research can either be applied (or action) research or fundamental (to basic or pure) research. *Applied research* aims at finding a solution for an immediate problem facing a society or an industrial/business organisation, whereas *fundamental research* is mainly concerned with generalisations and with the formulation of a theory. "Gathering knowledge for knowledge's sake is termed 'pure' or 'basic' research."[4] Research concerning some natural phenomenon or relating to pure mathematics are examples of fundamental research. Similarly, research studies, concerning human behaviour carried on with a view to make generalisations about human behaviour, are also examples of fundamental research, but research aimed at certain conclusions (say, a solution) facing a concrete social or business problem is an example of applied research. Research to identify social, economic or political trends that may affect a particular institution or the copy research (research to find out whether certain communications will be read and understood) or the marketing research or evaluation research are examples ofapplied research. Thus, the central aim of applied research is to discover a solution for some pressing practical problem, whereas basic research is directed towards finding information that has a broad base of applications and thus,adds to the already existing organized body of scientific knowledge.

(iii) ***Quantitative vs. Qualitative:*** Quantitative research is based on the measurement of quantity or amount. It is applicable to phenomena that can be expressed in terms of quantity. Qualitative research, on the other hand, is concerned with qualitative phenomenon, i.e., phenomena relating to or involving quality or kind. For instance, when we are interested in investigating the reasons for human behaviour (i.e., why people think or do certain things), we quite often talk of 'Motivation Research', an important type of qualitative research. This type of research aims at discovering the underlying motives and desires, using in depth interviews for the purpose. Other techniques of such research are word association tests, sentence completion tests, story completion tests and similar other projective techniques. Attitude or opinion research i.e., research designed to find out how people feel or what they think about a particular subject or institution is also qualitative research. Qualitative research is specially important in the behavioural sciences where the aim is to discover the underlying motives of human behaviour. Through such research we can analyse the various factors which motivate people to behave in a particular manner or which make people like or dislike a particular thing. It may be stated, however, that to apply qualitative research in practice is relatively a difficult job and therefore, while doing such research, one should seek guidance from experimental psychologists.

(iv) ***Conceptual vs. Empirical:*** Conceptual research is that related to some abstract idea(s) or theory. It is generally used by philosophers and thinkers to develop new concepts or to reinterpret existing ones. On the other hand, empirical research relies on experience or observation alone, often without due regard for system and theory. It is data-based research, coming up with conclusions which are capable of being verified by observation or experiment. We can also call it as experimental type of research. In such a research it is necessary to get at facts firsthand, at their source, and actively to go about doing certain things to stimulate the production of desired information. In such a research, the researcher must first provide himself with a working hypothesis or guess as to the probable results. He then works to get enough facts (data) to prove or disprove his hypothesis. He then sets up experimental designs which he thinks will manipulate the persons or the materials concerned so as to bring forth the desired information. Such research is thus characterised by the experimenter's control over the variables under study and his deliberate manipulation of one of them to study its effects. Empirical research is appropriate when proof is sought that certain variables affect other variables in some way. Evidence gathered through experiments or empirical studies is today considered to be the most powerful support possible for a given hypothesis.

(v) ***Some Other Types of Research:*** All other types of research are variations of one or more of the above stated approaches, based on either the purpose of research, or the time required to accomplish research, on the environment in which research is done, or on the basis of some other similar factor. Form the point of view of time, we can think of research either as *one-time research or longitudinal research*. In the former case the research is confined to a single time-period, whereas in the latter case the research is carried on over several time-periods. Research can be *field setting research or laboratory research or simulation research*, depending upon the environment in which it is to be carried out. Research can as well be understood as *clinical or diagnostic research*. Such research follow case-study methods or indepth approaches to reach the basic causal relations. Such studies usually go deep into the causes of things or events that interest us, using very small samples and very deep probing data gathering devices. The research may be *exploratory* or it may be formalized. The objective of exploratory research is the development of hypotheses rather than their testing, whereas formalized research studies are those with substantial structure and with specific hypotheses to be tested. *Historical research* is that which utilizes historical sources like documents, remains, etc. to study events or ideas of the past, including the philosophy of persons and groups at any remote point of time. Researchcan also be classified as *conclusion-oriented* and decision-oriented. While doing conclusion-oriented research, a researcher is free to pick up a problem, redesign the enquiry as he proceeds and is prepared to conceptualize as he wishes. Decision-oriented research is always for the need of a decision maker and the researcher in this case is not freeto embark upon research according to his own inclination. Operations research is an example of decision-oriented research since it is a scientific method of providing executive departments with a quantitative basis for decisions regarding operations under their control.

Process of Research:

The chart indicates that the research process consists of a number of closely related activities, as shown through I to VII. But such activities overlap continuously rather than following a strictly prescribed sequence. At times, the first step determines the nature of the last step to be undertaken. If subsequent procedures have not been taken into account in the early stages, serious difficulties may arise which may even prevent the completion of the study. One should remember that the various steps involved in a researchprocess are not mutually exclusive; nor they are separate and distinct. They do not necessarily follow each other in any specific order and the researcher has to be constantly anticipating at each step in the research process the requirements of the subsequent steps. However, the following order concerning various steps

provides a useful procedural guideline regarding the research process: (1) formulating the research problem; (2) extensive literature survey; (3) developing the hypothesis; (4) preparing the research design; (5) determining sample design; (6) collecting the data; (7) execution of the project; (8) analysis of data; (9) hypothesis testing; (10) generalisations and interpretation, and (11) preparation of the report or presentation of the results, i.e., formal write-up of conclusions reached.

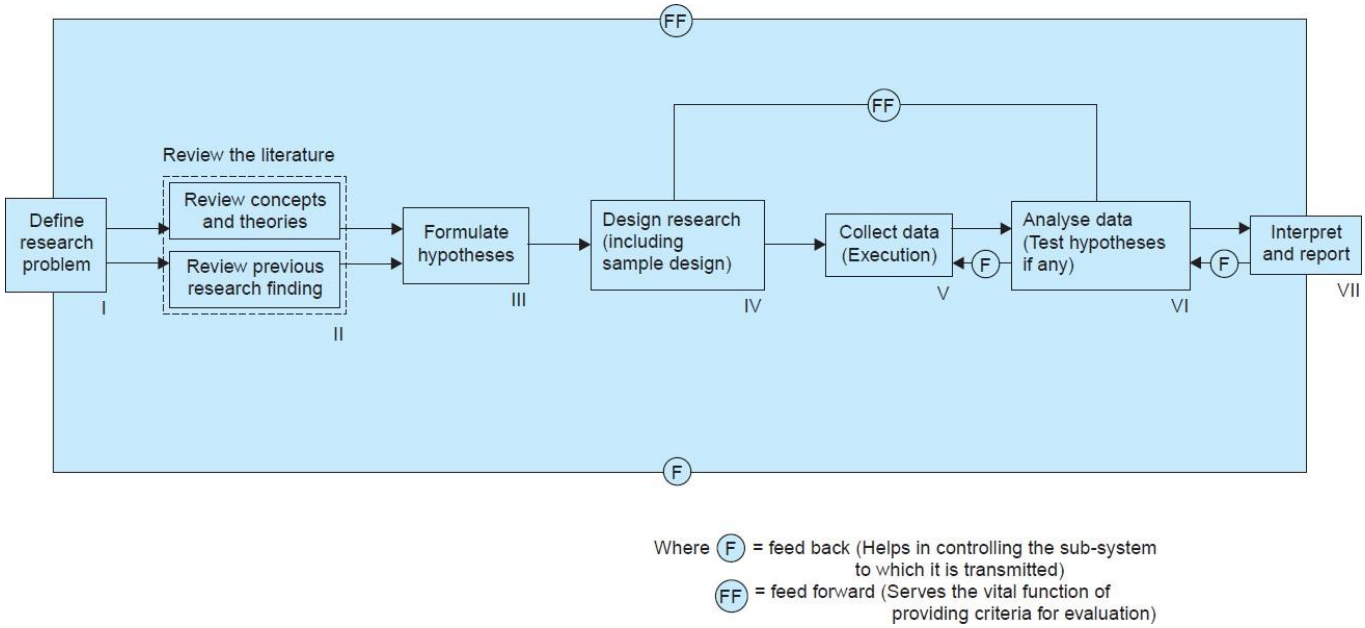


Figure: Process of Research

1. Problem identification: The first step in a research process is to identify the problem or opportunity. The problem may be about decrease in sales, increase in competition, expansion of market, etc.

2. Problem definition: The second step in a research process is to define the problem. In this stage, the researcher must understand the problem correctly. He must find out the scope of the problem, the type of information needed, etc. If the problem is not defined properly, then it will result in waste of time, money and resources.

3. Research design: The third step in a research process is to prepare research design. Research design is a plan for conducting a research. It guides the researcher in data collection. It gives proper direction to the research. There are three types of research designs:

1. Exploratory research,

2. Descriptive Research and

3. Experimental Research.

All three types are used for marketing research.

4. **Determining data needs:** The fourth step in a research process is to determine the data needs. The researcher must consider the following issues:

1. Whether to use primary data or secondary data or both.

2. The accuracy and reliability of the data.

3. The availability of accurate and reliable data.

4. The cost and time required to collect the data.

5. **Determining data sources:** The fifth step in a research process is to determine the data sources. The researcher decides the sources of collecting data. The two main sources are secondary data and primary data. The researcher first collects secondary data. This is because it is easily available and less costly. It is collected by Desk Research. Desk Research can be internal for e.g. collected from company's records or external i.e. acquired from libraries, trade journals, government sources, etc. If the secondary data is not sufficient to solve the marketing problem, then primary data is wheeled. Collecting primary data is very costly and time consuming. It can be collected by using survey methods, i.e. by doing personal interviews, telephone interviews and mail surveys. It can also be collected by using observational method and experimentation method. So in this step the researcher decides what source and what method to use for collecting data.

6. **Sampling design:** The sixth step in a research process is of sampling design. The Researchers has limited time and other resources. So he cannot contact the total population. That is, he cannot collect information from all the people in the market. Therefore, he selects few persons from the population. These handful persons are called sample respondent. They are considered to represent the total population. The researcher collects data from the sample respondents. Sampling helps to save time, efforts and cost. It is used to collect primary data. The researcher has to decide about method of sampling, the size-of-sample, etc.

7. **Designing questionnaire:** The seventh step in a research process is of designing a questionnaire. In this stage, primary data is collected with the help of a questionnaire. So the researcher has to prepare a questionnaire. A questionnaire is a list of questions. These questions are asked to the respondents for collecting data. The questionnaire must be suitable so that the require data is collected easily, quickly and correctly. It is used for conducting person interview, telephone interviews and mail survey. The researcher must decide about the type of the information required, the type of questioned to be asked, the wordings of the questionnaire, its order, etc.

8. **Field staff selection:** The eighth step in a research process is of selecting field staff. After preparing the questionnaire, the researcher selects field interviewers. The field interviewers collect information from the respondents. They must be property trained. Students of psychology and statistics are good for this job.

9. **Collection and processing of data:** The ninth step in a research process is of collection and processing of data. In this stage, the data is collected from the respondents. The questionnaire is used for collecting data. In case of mail surveys, the questionnaire is sent to the respondents by post. In case of telephone interviews, the data is collected through telephone. In case of personal interviews, the data is collected by the field interviewers. The researcher can also use observation method and experimentation method for collecting data. The data collected must be reliable and complete. It must also be collected quickly. Secondary data is also collected. The data collected is raw. It cannot be used directly. It has to be processed and organized neatly. That is, the data must be edited, coded, classified and tabulated. Editing helps to remove the unwanted data. Coding, classification and tabulation make the data ready for analysis and interpretation.

10. **Analysis and interpretation of data:** The tenth step in a research process is of analysis and interpretation of data. In this stage, the researcher analyzes and interprets the data. That is, he studies the data very careful and draws conclusions from it. These conclusions are then used to solve the marketing problem.

11. **Project reporting:** The eleventh step in a research process is to prepare a project report. In this stage, the researcher prepares the final research report. This report contains a title of the report, method used,

findings, conclusions and suggestions about how to solve the marketing problem. The language of the report must not be very difficult. The report must be submitted to the marketing executives for recommendations and implementation.

Finding and Solvig a worthwhile problem:

Research Problem: A research problem is a statement about an area of concern, a condition to be improved, a difficulty to be eliminated, or a troubling question that exists in scholarly literature, in theory, or in practice that points to the need for meaningful understanding and deliberate investigation.

What is research problem statement?

A problem statement is the description of an issue currently existing which needs to be addressed. It provides the context for the research study and generates the questions which the research aims to answer. The statement of the problem is the focal point of any research.

How do you identify the problem?

The main steps in identification & formulation of research problem are:

1. Specify the Research Objectives

A clear statement of objectives will help you develop effective research. It will help the decision makers evaluate your project. It's critical that you have manageable objectives. (Two or three clear goals will help to keep your research project focused and relevant.)

2. Review the Environment or Context of the Research Problem

As a marketing researcher, you must work closely with your team. This will help you determine whether the findings of your project will produce enough information to be worth the cost. In order to do this, you have to identify the environmental variables that will affect the research project.

3. Explore the Nature of the Problem

Research problems range from simple to complex, depending on the number of variables and the nature of their relationship. If you understand the nature of the problem as a researcher, you will be able to better develop a solution for the problem. To help you understand all dimensions, you might want to

consider focus groups of consumers, sales people, managers, or professionals to provide what is sometimes much needed insight.

4. Define the Variable Relationships

Marketing plans often focus on creating a sequence of behaviors that occur over time, as in the adoption of a new package design, or the introduction of a new product. Such programs create a commitment to follow some behavioral pattern in the future. Studying such a process involves:

- Determining which variables affect the solution to the problem.
- Determining the degree to which each variable can be controlled.
- Determining the functional relationships between the variables and which variables are critical
- to the solution of the problem.

During the problem formulation stage, you will want to generate and consider as many courses of action and variable relationships as possible.

5. The Consequences of Alternative Courses of Action

There are always consequences to any course of action. Anticipating and communicating the possible outcomes of various courses of action is a primary responsibility in the research process.

Necessity of Defining a Research Problem:

The problem to be researched needs to be described unambiguously as that will help you to discriminate useful data from the unrelated ones. A proper formulation of research problem will allow the investigator to be on the track in contrast to an ill-defined problem may possibly create difficulties.

Questions like: What data are to be gathered? What attributes of data are appropriate and need to be analyzed? What relations should be investigated. What methods should be employed for the purpose? aswell as other questions turn up in the head of the investigator who can well plan his strategy and find solutions to these kinds of questions only when the research problem has been well defined.

Therefore, defining the problem accurately is a necessity for any research and is a step of the highest value. In fact, formulation of a problem is often vital than its solution. It is only on thoroughly describing the

problem that we can work out the research design and can efficiently proceed all the consequential steps needed while doing research.

Important Points to Keep in Mind while Defining the Research Problem

1. The correct question needs to be addressed if research is to help decision makers. A right answer to the wrong question leads either to bad advice or to no advice.

2. Usually in problem we have an inclination to rationalize and defend our actions once we have started upon a specific research plan. The perfect time to examine and think about alternative techniques is in the planning stage. If it is completed unnecessary expense of false start and redoing work may be prevented.

3. An excellent beginning in problem definition is to ask what the decision maker want to know if the requested information can be gathered without error and without expense.

4. Another excellent rule to follow is "**Never settle on a specific strategy**" without developing and taking into consideration at least one alternate option".

5. The problem definition stage of research is the determination and structuring of the decision maker's question. It should be the decision maker's question and not the researcher's question.

6. What decision do you face? Unless you have decision to make, there isn't any research problem.

7. What are the alternatives? In case there are no options to choose, once again there is absolutely no research problem.

8. What are the factors for selecting the best alternative? Unless you have criteria for evaluation, again there's no problem.

9. The researcher should stay away from the acceptance of the superficial and the obvious. Frequently we all hear that a problem clearly expressed is a problem half solved. This statement indicates the *necessity of defining a research problem in research methodology*. This actually also results in a smoother progress on all the following steps which are needed for finishing a research project.

Here are seven-steps for an effective problem-solving process.

1. Identify the issues. Be clear about what the problem is.

2. Understand everyone's interests.

3. List the possible solutions (options).

4. Evaluate the options.

5. Select an option or options.

6. Document the agreement(s).

7. Agree on contingencies, monitoring, and evaluation.

Ethics in Engineering Research:

Ethics generally refers to a set of rules distinguishing acceptable and unacceptable conduct, distinguishing right from wrong, or wise aphorisms like the sayings of Chanakya. Most people learn such norms in their formative years, but moral development continues through different stages of growth. Although everyone recognizes some common ethical norms, but there is difference in interpretation and application. Ethical principles can be used for evaluation, proposition, or interpretation of laws. Although ethics are not laws, but laws often follow ethics because ethics are our shared values.

Ethics are norms or standards of behavior that guide moral choices about our behavior and our relationship with others. As in other aspects of business, all parties in research should exhibit ethical behavior. The goal of ethics in research is to ensure that no one is harmed or suffered adverse consequences from research activities. However, unethical activities are pervasive and include violating non-disclosure agreement, breaking respondent confidentiality, misrepresenting results, deceiving people, invoicing irregularities, avoiding legal liability, and more.

- As research is designed, several ethical considerations must be balanced:
- Protect the rights of the participant or subject.
- Ensure the sponsor receives ethically conducted and reported research
- Follow ethical standards when designing research
- Protect the safety of the researcher and team
- In general, research must be designed so a respondent does not suffer physical harm, discomfort, pain, embarrassment, or loss of privacy. Begin data collection by explaining to respondent the benefits expected from the research.

- Explain that their rights and well-being will be adequately protected and say how that will be done. Be certain that interviewers obtain in the inform consent of the respondent. The use of deception is questionable; when it is used, debrief any respondent who has been deceived.

Ethics in Engineering Research Practice:

- **Honesty:** Strive for honesty in all scientific communications. Honestly report data, results, methods and procedures, and publication status. Do not fabricate, falsify, or misrepresent data. Do not deceive colleagues, research sponsors, or the public.
- **Objectivity:** Strive to avoid bias in experimental design, data analysis, data interpretation, peer review, personnel decisions, grant writing, expert testimony, and other aspects of research where objectivity is expected or required. Avoid or minimize bias or self-deception. Disclose personal or financial interests that may affect research.
- **Integrity:** Keep your promises and agreements; act with sincerity; strive for consistency of thought and action.
- **Carefulness:** Avoid careless errors and negligence; carefully and critically examine your own work and the work of your peers. Keep good records of research activities, such as data collection, research design, and correspondence with agencies or journals.
- **Openness:** Share data, results, ideas, tools, resources. Be open to criticism and new ideas. Respect for Intellectual Property: Honor patents, copyrights, and other forms of intellectual property. Do not use unpublished data, methods, or results without permission. Give proper acknowledgement or credit for all contributions to research. Never plagiarize.
- **Confidentiality:** Protect confidential communications, such as papers or grants submitted for publication, personnel records, trade or military secrets, and patient records.
- **Responsible Publication:** Publish in order to advance research and scholarship, not to advance just your own career. Avoid wasteful and duplicative publication.
- **Responsible Mentoring:** Help to educate, mentor, and advise students. Promote their welfare and allow them to make their own decisions.

- **Respect for colleagues:** Respect your colleagues and treat them fairly.
- **Social Responsibility:** Strive to promote social good and prevent or mitigate social harms through research, public education, and advocacy.
- **Non-Discrimination:** Avoid discrimination against colleagues or students on the basis of sex, race, ethnicity, or other factors not related to scientific competence and integrity
- **Competence:** Maintain and improve your own professional competence and expertise through lifelong education and learning; take steps to promote competence in science as a whole.
- **Legality:** Know and obey relevant laws and institutional and governmental policies.
- **Animal Care:** Show proper respect and care for animals when using them in research. Do not conduct unnecessary or poorly designed animal experiments.
- **Human Subjects Protection:** When conducting research on human subjects, minimize harms and risks and maximize benefits; respect human dignity, privacy, and autonomy; take special precautions with vulnerable populations; and strive to distribute the benefits and burdens of research fairly.

Research Misconduct:

Research Misconduct can be characterized as actions or questionable research practices that fall short of the standards of ethics, research and scholarship required to ensure that the integrity of research is upheld. It can cause harm to people and the environment, wastes resources, undermines the research record and damages the credibility of research.

It is often defined by 'falsification, fabrication and plagiarism' and can include making up data or results, incorrectly attributing authorship, gift authorship, manipulating research materials, equipment, or processes, or changing or omitting data, graphs, images or results.

Questionable research practices encompass a much wider group of misdemeanours, poor research design and other unhealthy practices. Some of these may be deployed in ignorance of the potential consequences for the integrity of the research rather than attempt to mislead – this is often referred to as 'sloppy science'.

What are the consequences of research misconduct?

The consequences of research misconduct can be severe including preventable illness or the loss of human life due to misinformation in the literature or continued citing of retracted work. It can also result in wasted resources, both human and financial, when newer research or work is based on previous flawed or fraudulent research.

Funding agencies often require that cases of research misconduct are reported to them which can be damaging to the careers of those who commit misconduct and there is a financial cost to the institution in investigating allegations. The retraction of papers and reputational risks of misconduct can be damaging to the research careers of those who commit misconduct.

Types of research misconduct:

There are different types of research misconduct or scientific misconduct and unethical practices in research. The most serious ethical infractions are fabrication, falsification, and plagiarism in research. Some of the most common types of research misconduct have been detailed below.

- ***Fabrication:*** This refers to the practice of making up data without having done the required research. Research misconduct covers not only the act of fabrication, but also the sharing, discussing, or publishing of this fabricated data or results.
- ***Falsification:*** This type of scientific misconduct involves the wilful manipulation of data, materials, processes, or equipment to arrive at a predefined conclusion. One such example would be selectively omitting or changing data, which results in the erroneous representation of research results.
- ***Plagiarism:*** This is one of the most common types of scientific misconduct, and involves using another person's ideas, content, writing, processes, or results without giving due credit. This also includes ***self-plagiarism***, which occurs when you replicate your own writings or ideas from previously published research without providing proper credit.

- ***Authorship:*** This type of scientific misconduct in research includes attempts to assign false authorships without adequate contribution to research, mentioning authors without their consent, or failing to include authors who are original contributors. Naming authors in the wrong order or incorrectly is also considered unethical.
- ***Conflicts of interest:*** This can be classified under general scientific misconduct and involves lapses by researchers in declaring any conflict of interests in their research work. These conflicts of interest may be financial, personal, and professional and need to be reported appropriately to avoid any ethical issues.
- ***Approvals:*** One of the most important aspect of research that involves human or animal subjects is adhering to all the ethical approvals and legal guidelines. Non-compliance with this ethical mandate is considered a serious type of research misconduct.

5 Reasons for committing research misconduct

Over time there have been varied reasons for researchers to succumb to scientific misconduct. Let us look at 5 reasons for committing research misconduct.

- ***Career pressures*****:** An important factor often associated with research misconduct is the unduepressure researchers face. They need to conduct original research in a fast-paced environment, publish frequently in peer reviewed journals, and procure funding for research projects to advance their research career. This along with the need to juggle multiple responsibilities against tightdeadlines create undue stress to succeed at any cost, leading to a lack of care or even deliberate research misconduct.
- ***Researcher's personal psychology:*** Some researchers may be overly driven by a desire to quickly attain a strong professional reputation or even financial gains, which could push them to research misconduct.
- ***Lack of appropriate training and skills:*** The lack of training on the best practices and ethical guidelines to be followed as researchers is another reason for research misconduct. Poor awareness and understanding on these issues often lead to unethical conduct in research.

- ***Insufficient supervision or mentoring:*** Related to the point above, this relates to situations where researchers, especially early career researchers, fail to receive sufficient and appropriate support from immediate supervisors or from their affiliated institution. A lack of oversight and guidance may knowingly or unknowingly lead to research misconduct.
- ***Inadequate knowledge:*** Scientific misconduct can occur if the researcher does not have sufficient knowledge of the topic/subject or on research best practices. Carelessness when conducting research and reporting it are also considered research misconduct.

Ethical issues related to Authorship:

Criteria of Authorship:

Academic life revolves around publications. The publication adds to the credibility of the research and brings fame and recognition. An author is an individual who fulfills enlisted criteria collectively:

(1) substantial contributions to conception and design

(2) acquisition of data, or analysis and interpretation of data

(3) drafting the article or revising it critically for important intellectual content and

(4) final approval of the version to be published.

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